



02451 Introduction to Machine Learning

Week 5: Bayesian inference and linear models

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3 March 2026

DTU Compute, Technical University of Denmark

Today

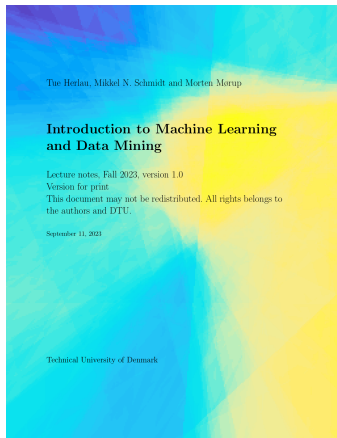
Feedback Groups of the day:

Luca Aiello, Narin Ali Ehyaii, Dima Al-Khamisi, Olivia Lior Bareket, Alexander Baumkirchner, Rasmus Søgård Bertelsen, Christian Bukkjær, Andreas Stefan Bjørn Bundgaard, Alexander Harry Mallock Cadell, Rune Christensen, Jesper Minet Christensen, Lorenzo Cuoci, Hugo Daniel Da Silva Guiomar, Villads Dehli, Laurits Enemark Due, Haris Fazlagic, Carlos Ricardo Harris, Oscar Illerup-Gunnersen, Una Mattý Jensdóttir, Oskar Skøtt Johansen, Oliver Kabel, Rasmus Hjort Kleffel, Adam Blichert Klovborg, William Mac Larsen, David Borgbjærg Madsby, Pablo Manjon Delgado, Aisha Farah Mohamud, Viktor Munk, Mathias Hoberg Nielsen, Leah Halkjær Olesen, Magnus Starkad Ottosen, Sally Holst Riisberg, Nico Schaul, Youssef Taouzlak, Miguel André Marques Trindade, Freja Venzel, Laura Ruby Wimmelmann.

Reading/homework material:

Chapter 5, 6, 8

P8.1, P8.2



Lecture Schedule

- 1 Introduction, data, and visualization
3 February: C1, C2, C7
- 2 Summary statistics, similarity, and nearest neighbors
10 February: C4, C12
- 3 Computational linear algebra and PCA
17 February: C3
- 4 Probability theory, Bayes and Naive Bayes
24 February: C5, C6, C13 (Assignment 1-3 due on 26 February)
- 5 **Bayesian inference and linear models**
3 March: C5, C6, C8
- 6 Decision trees, overfitting and cross-validation
10 March: C9, C10
- 7 Performance evaluation and AUC
17 March: C11, C16 (Assignment 4-6 due on 19 March)
- 8 Artificial Neural Networks and Optimization
24 March: C15
- 9 Bias/Variance and ensemble methods
7 April: C14, 17
- 10 K-means and hierarchical clustering
14 April: C18 (Assignment 7-9 due on 16 April)
- 11 Mixture models and density estimation
21 April: C19, C20
- 12 Representation learning
28 April: Material provided separately
- 13 Recap and discussion of the exam
5 May: C1-C20 (Assignment 10-12 due on 7 May)

Online help: Piazza

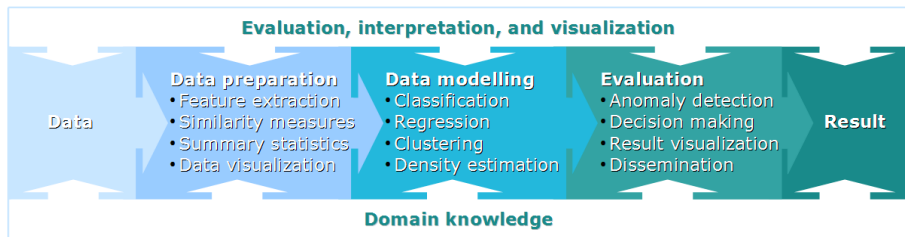
Videos of lectures: <https://panopto.dtu.dk>

Streaming of lectures: Zoom (link on DTU Learn)

Practicalities and announcements

- Old exam sets.
- Exercise/assignments - remember the survey!

Learning Objectives



Learning Objectives

- Understand the maximum likelihood and maximum a posteriori principles
- Derive cost-functions from likelihood functions using Bayes theorem
- Apply and interpret linear regression and logistic regression

Plan for today:

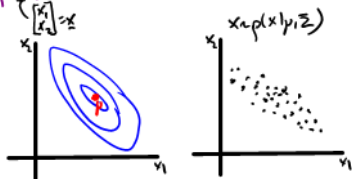
- Lecture 5 (13:00 – ~15:00)
 - Basic principles of Bayesian inference
 - Learning via statistical inference
 - Linear models for regression (*linear regression*)
 - Linear models for classification (*logistic regression*)
- Exercises (15:00 – 17:00)

Recap lecture 4

- Binary events and Bayes rule.
- Discrete probabilities (e.g., Bernoulli).
- Continuous densities (e.g., Gaussian).
- (Naïve) Bayes example.

Multivariate Gaussian Distribution

$$p(\mathbf{x}|\mu, \Sigma) = \mathcal{N}(\mathbf{x}|\mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x}-\mu)^T \Sigma^{-1}(\mathbf{x}-\mu)\right\}$$



Discrete probabilities
- Bernoulli dist.

$$p(b|\theta) = \theta^b (1-\theta)^{1-b}$$

$$p(b=1) = \theta$$

$$p(b=0) = (1-\theta)$$

Binary events

$$p(A|B)p(B) = p(B|A)p(A) = p(A,B)$$

$$p(B) = \sum_{A \in \{1,3\}} p(A=0, B), \quad \sum_{A \in \{1,3\}} p(A=1) = 1$$

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)}$$

binary events

Random variables

		orange	green	red
U	1	3	4	
V	2	2	4	
	3	5	8	

$A = \begin{cases} 1, \text{select orange} \\ 0, \text{select apple} \end{cases}$

$B = \begin{cases} 1, \text{select from blue bowl} \\ 0, \text{select from green bowl} \end{cases}$

Bayes & Naive Bayes Classifier

$$p(y=0|x_1, x_2) = \frac{p(x_1, x_2|y=0)p(y=0)}{p(x_1, x_2)}$$

$$p(x_1, x_2|y=0)p(y=0) + p(x_1, x_2|y=1)p(y=1)$$

Naive Bayes Assumption

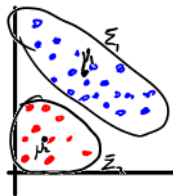
$$p(x_1, x_2|y) = p(x_1|y)p(x_2|y)$$

If MVN

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \text{cov}(x_1, x_2) \\ \text{cov}(x_2, x_1) & \sigma_2^2 \end{bmatrix}$$

$$p(x_1|y) = \mathcal{N}(x_1|\mu_1, \sigma_1^2)$$

$$p(x_2|y) = \mathcal{N}(x_2|\mu_2, \sigma_2^2)$$



Bayesian inference

Discrete Example

Coin flips: $\{H, T, T, H, T, H, H, T\}$

Bernouli $p(b|\theta) = \theta^b (1-\theta)^{1-b}$

$b = \begin{cases} 1, & \text{head} \\ 0, & \text{tails} \end{cases}$

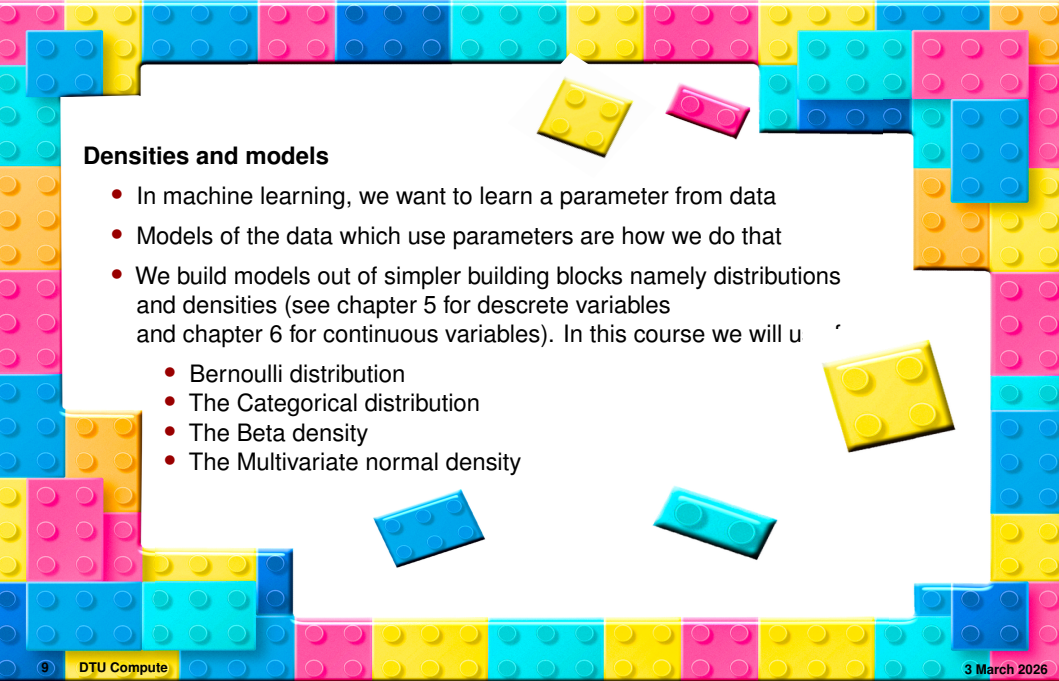
- What is the $\hat{\theta}$ that we can estimate from our observations?
- What is the $\hat{\theta}$ if $\{H, H, H, H, H, T\}$?

Continuous Example

Heights of students in the room? $\{1.72, 1.85, 1.82, 1.77, 1.65, 1.91, \dots\}$

Gaussian density $p(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}\right)$

- Estimate $\hat{\mu}$ and $\hat{\sigma}^2$ from data
- if all students today above 1.95?
- Does it make sense?



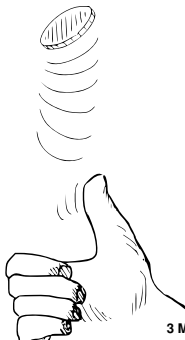
Densities and models

- In machine learning, we want to learn a parameter from data
- Models of the data which use parameters are how we do that
- We build models out of simpler building blocks namely distributions and densities (see chapter 5 for discrete variables and chapter 6 for continuous variables). In this course we will use
 - Bernoulli distribution
 - The Categorical distribution
 - The Beta density
 - The Multivariate normal density

Reminder: The Bernoulli distribution

- Let $b = 0, 1$ denote a binary event.
- For instance,
 - $b = 0$ corresponds to heads, and $b = 1$ to tails, or
 - $b = 0$ corresponds to a person being ill, and $b = 1$ that a person is well.
- The probability of b is expressed using a parameter θ in the unit interval $[0, 1]$

Bernoulli distribution: $p(b|\theta) = \theta^b(1 - \theta)^{1-b}.$



The Bernoulli distribution and repeated events

- Suppose we observe a sequence $b_1, \dots, b_N$ of Bernoulli (binary) events.
- For instance, for N patients we record whether person 1 is ill or well ($b_1 = 0$ or $b_1 = 1$) and up to whether patient N is ill or well ($b_N = 0$ or $b_N = 1$).
- When we **know** θ (the chance a person is well or ill), we **assume** that the events are **independent**

Bernoulli distribution: $p(b|\theta) = \theta^b(1-\theta)^{1-b}$.

Likelihood

*Probability of observed
sequence of $b_1, b_2, \dots, b_N$
given value of θ*

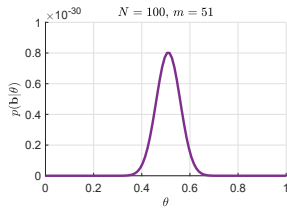
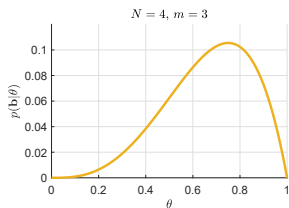
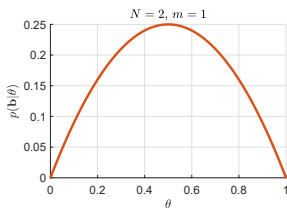
$$\begin{aligned} p(b_1, \dots, b_N | \theta) &= \prod_{i=1}^N p(b_i | \theta) \\ &= \prod_{i=1}^N \theta^{b_i} (1 - \theta)^{1-b_i} \end{aligned}$$

$$= \theta^{\sum_{i=1}^N b_i} (1 - \theta)^{N - \sum_{i=1}^N b_i}$$

$$= \theta^m (1 - \theta)^{N-m}, \quad m = b_1 + b_2 + \dots + b_N$$

*Recall for cond. indep.
 $p(x, y | z) = p(x | z) p(y | z)$*

Learning principle: Maximum likelihood estimation (MLE)



$$p(b_1, \dots, b_N | \theta) = \theta^m (1 - \theta)^{N-m}$$

An idea for selecting θ^* is **Maximum Likelihood Estimation (MLE)**

$$\theta^* = \arg \max_{\theta} p(b_1, \dots, b_N | \theta)$$

The value of θ according to which the data is most plausible (i.e., a point estimate)

plug into
Bayes and spec.
prior

Maximum
Likelihood
 $\theta^* = \operatorname{argmax}_{\theta} p(x|\theta)$

Bayesian inference
$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)}$$


only point estimate
but keep prior

MAP est.
$$\theta^* = \operatorname{argmax}_{\theta} p(\theta|x)$$
$$= \operatorname{argmax}_{\theta} p(x|\theta)p(\theta)$$


Drop prior $p(\theta)$

Probabilities, densities and learning

- Your friend just got a dog.
- The dog can either be in the doghouse or outside
- The first four times you come by the dog is in the doghouse
- What is the chance the dog is in the doghouse tomorrow?



- Your friend buys a coin.
- The coin can either come up heads or tails
- The first four times you flip the coin it comes up tails
- What is the chance the coin comes up tails in the next flip?

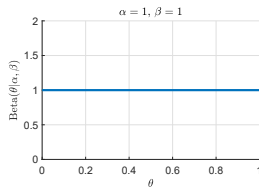
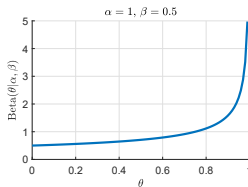
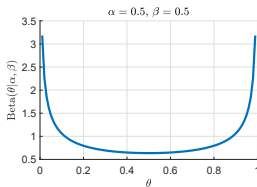


Intuition tells us the answers are different, but the situation seems similar...

Beta distribution

Suppose θ is defined on the unit interval $\theta \in [0, 1]$

$$\text{Beta density: } p(\theta \mid \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$$

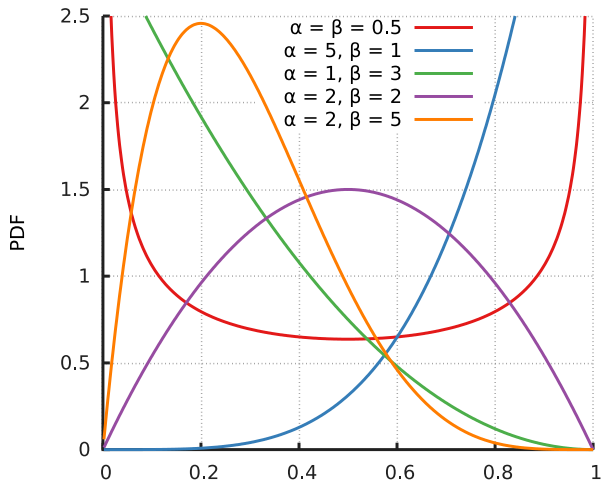


$\alpha, \beta > 0$ are related to the mean and variance

$$\mathbb{E}_{p(\theta|\alpha,\beta)}[\theta] = \frac{\alpha}{\alpha + \beta}$$

$$\text{Var}_{p(\theta|\alpha,\beta)}[\theta] = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Beta distribution



Picture from Wikipedia

The posterior distribution

- A magic coin is a coin that comes up heads with probability θ
 - Suppose $b = 1$ is the event the coin land heads
 - $b = 0$ is the event the coin land tails
- For a sequence of N flips $b_1, b_2, \dots, b_N$

$$p(b_1, \dots, b_N) = \theta^m (1 - \theta)^{N-m}, \quad m = b_1 + b_2 + \dots + b_N$$

- **What is θ ?** Answer: **Use Bayes' Theorem!**

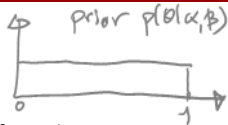
"posterior" $\rightarrow$ $p(\theta|\mathbf{b}) = \frac{p(\mathbf{b}|\theta)p(\theta)}{p(\mathbf{b})} = \frac{p(\mathbf{b}|\theta)p(\theta)}{\int_0^1 p(\mathbf{b}|\theta')p(\theta')d\theta'}$
Likelihood $\rightarrow$ $p(\mathbf{b}|\theta)$
prior $\rightarrow$ $p(\theta)$

"evidence" $\rightarrow$ $p(\mathbf{b})$

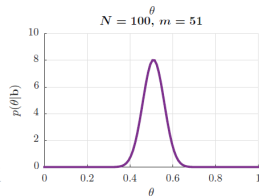
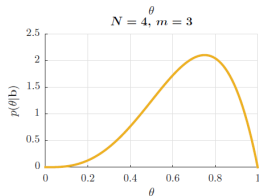
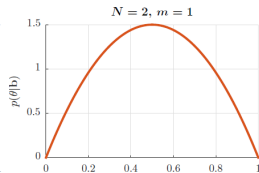
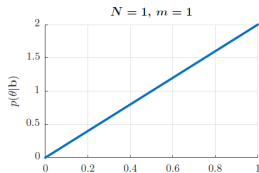
- Assume $p(\theta) = p(\theta|\alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$

$$\begin{aligned}
 p(\theta|\mathbf{b}, \alpha, \beta) &= \frac{\theta^m (1 - \theta)^{N-m} \times \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}}{\int \theta'^m (1 - \theta')^{N-m} \times \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta'^{\alpha-1} (1 - \theta')^{\beta-1} d\theta'} \\
 &= \frac{\Gamma(\alpha + \beta + N)}{\Gamma(\alpha + m)\Gamma(\beta + N - m)} \theta^{\alpha+m-1} (1 - \theta)^{\beta+N-m-1}
 \end{aligned}$$

Example: $\alpha = \beta = 1$



$$\begin{aligned} p(\theta|\mathbf{b}, \alpha, \beta) &= \frac{\Gamma(\alpha + \beta + N)}{\Gamma(\alpha + m)\Gamma(\beta + N - m)} \theta^{\alpha+m-1} (1 - \theta)^{\beta+N-m-1} \\ &= \frac{(N+1)!}{m!(N-m)!} \theta^m (1 - \theta)^{N-m} \end{aligned}$$



Dogs and coins



- Your friend just got a dog.
- The dog can either be in the doghouse or outside
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Prior

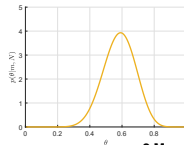
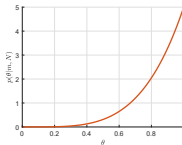
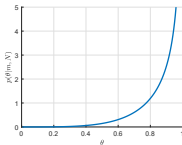
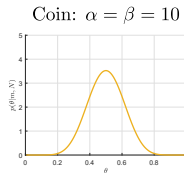
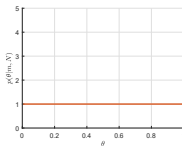
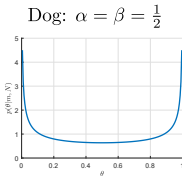
$$p(\theta \mid \alpha, \beta) =$$

Likelihood

$$p(m = 4, N = 4 \mid \theta) = \theta^m (1 - \theta)^{N-m} = \theta^4 (1 - \theta)^{4-4} = \theta^4$$

Posterior distribution

$$p(\theta \mid m = 4, N = 4) = \frac{p(N, m \mid \theta) p(\theta \mid \alpha, \beta)}{p(N, m)}$$



Learning principle: Maximum a posteriori (MAP)

- **Another idea:** Select θ which is "most probable"

$$\theta^* = \arg \max_{\theta} p(\theta|m, N) = \arg \max_{\theta} \left[\frac{p(m, N|\theta)p(\theta|\alpha, \beta)}{p(m, N)} \right]$$

- Use that $\arg \max_x f(x) = \arg \min_x [-\log f(x)]$ if $f(x) > 0$:

$$\theta^* = \arg \min_{\theta} \left[-\log \frac{p(m, N|\theta)p(\theta|\alpha, \beta)}{p(m, N)} \right]$$

(likelihood)
 $p(m, N|\theta) = \theta^m (1 - \theta)^{N-m}$

(prior)
 $p(\theta|\alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1}$

- All in all:

$$\theta^* = \arg \min_{\theta} E(\theta), \quad E(\theta) = -\log p(m, N|\theta) - \log p(\theta|\alpha, \beta)$$

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- **Another idea:** Select θ which is "most probable"

$$\theta^* = \arg \max_{\theta} p(\theta|m, N) = \arg \max_{\theta} \left[\frac{p(m, N|\theta)p(\theta|\alpha, \beta)}{p(m, N)} \right]$$

- Use that $\arg \max_x f(x) = \arg \min_x [-\log f(x)]$ if $f(x) > 0$:

$$\begin{aligned}\theta^* &= \arg \min_{\theta} \left[-\log \frac{p(m, N|\theta)p(\theta|\alpha, \beta)}{p(m, N)} \right] \\ &= \arg \min_{\theta} [-\log p(m, N|\theta) - \log p(\theta|\alpha, \beta) + \cancel{\log p(m, N)}] \\ &= \arg \min_{\theta} [-\log p(m, N|\theta) - \log p(\theta|\alpha, \beta)]\end{aligned}$$

- All in all:

$$\theta^* = \arg \min_{\theta} E(\theta), \quad E(\theta) = -\log p(m, N|\theta) - \log p(\theta|\alpha, \beta)$$

General Bayesian inference formulation

- Consider some data $\mathbf{X} = \mathbf{x}_1, \dots, \mathbf{x}_N$ and $\mathbf{y} = y_1, \dots, y_N$
- Suppose we think $\mathbf{x}_i$ relates to y_i by some parameters $\mathbf{w}$
- **Assume**

$$p(\mathbf{y}|\mathbf{X}, \mathbf{w}) = \prod_{i=1}^N p(y_i|\mathbf{x}_i, \mathbf{w}), \quad p(\mathbf{w}|\mathbf{X}) = p(\mathbf{w})$$

- Then

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{X}, \mathbf{w})p(\mathbf{w}|\mathbf{X})}{p(\mathbf{y}|\mathbf{X})}$$

- The following are equivalent:

$$\text{Maximize: } \mathbf{w}^* = \arg \max_{\mathbf{w}} p(\mathbf{w}|\mathbf{X}, \mathbf{y})$$

$$\text{Minimize: } \mathbf{w}^* = \arg \min_{\mathbf{w}} E(\mathbf{w}), \quad E(\mathbf{w}) = \frac{1}{N} \left[\sum_{i=1}^N -\log p(y_i|\mathbf{x}_i, \mathbf{w}) - \log p(\mathbf{w}) \right]$$

- All we need is a likelihood (**usually pretty simple**) and a prior (**can be omitted**) and we have a machine-learning method.
 - **Pro**: Easy, conceptually simple, efficient
 - **Con**: Can sometimes give spurious results (overfit)

$$f(x; w) = w_0 + \sum_{k=1}^K w_k x_k = w^T x$$

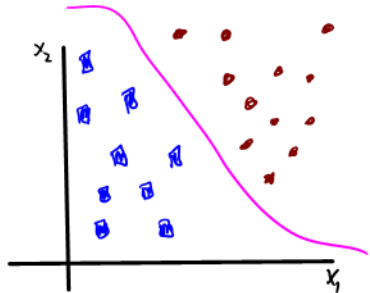
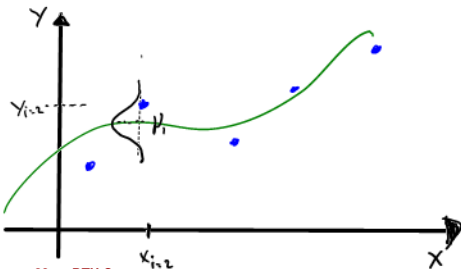
Linear models

(regression and classification)

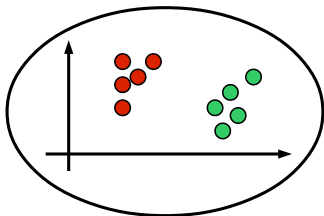
$$p(y = \square | w, x_i) = \theta_i$$

$$\theta_i = \sigma(w^T x_i)$$

$$p(y_i | p_i = f(x_i; w), \sigma) = \mathcal{N}(y_i | \mu_i = f(x_i; w), \sigma^2)$$



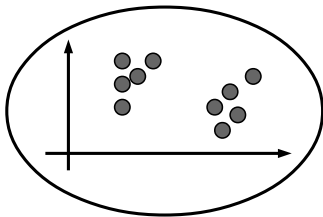
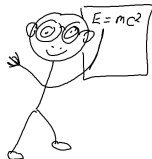
Supervised and Unsupervised learning



Supervised Learning

Input data $\mathbf{x}_n$ and output y_n

(Generalize from known examples)



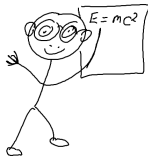
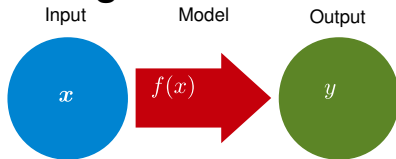
Unsupervised Learning

Input data $\mathbf{x}_n$ alone

(Exploratory analysis)



Supervised learning



- **Data**

- Inputs and outputs (*this is what we are given*)

$$\{\mathbf{x}_i, y_i\}_{i=1}^N$$

- **Model**

- Function with parameters w that maps inputs to outputs (*what we are trying to determine*)

$$f(\mathbf{x}; w)$$

- **Cost function**

- Dissimilarity measure between observation and prediction (*how we tell if a model is good or bad*)

$$d(y, f(\mathbf{x}; w))$$

- **Types** of supervised learning (in this course)

- Regression: Continuous output y
- Classification: Discrete output y

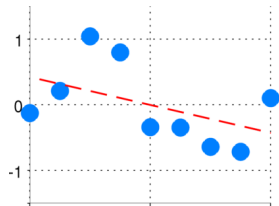
Regression

- **Definition:** Learning a function, $f(x)$, that maps a data object to a continuous-valued output
- **Why Regression?**
 - Descriptive modeling
 - Explain / understand the relation between attributes and continuous-valued output
 - Predictive modeling
 - Predict the output value of a new data object
- **Methods:** Regression trees, kNN regression, linear regression (today). *Later:* Regularized linear regression/ridge regression, neural networks, ensembles.

Linear models for regression

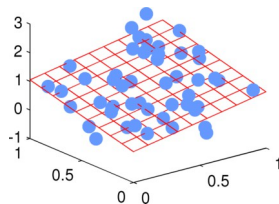
- 1-dimensional inputs

$$f(x) = w_0 + w_1x$$



- 2-dimensional inputs

$$f(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2$$



- M -dimensional inputs

$$f(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + \cdots + w_Mx_M$$

Linear models for regression

$f(x)$ short hand for $f(x;w)$
 $f_w(x)$

- K-dimensional inputs

$$f(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + \cdots + w_Mx_M$$

- Non-linearly transformed inputs

$$f(x) = w_0 + w_1x + w_2x^2 + \cdots + w_Kx^K$$

$$f(x) = w_0 + w_1\sin(x) + w_2\cos(x)$$

$$f(\mathbf{x}) = w_0 + w_1\sin(x_1) + w_2x_2^3$$

$$f(x;w) = \sum_{k=1}^K w_k \exp\left\{-\frac{1}{2\sigma^2}\|x - \mu_k\|^2\right\}$$

★ Allows non-linear functions in the input, x , but the model is still linear in the weights!

Linear models for regression

- K-dimensional inputs

$$f(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + \cdots + w_Mx_M$$

- Non-linearly transformed inputs

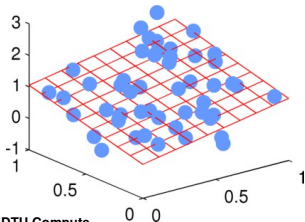
$$f(x) = w_0 + w_1x + w_2x^2 + \cdots + w_Kx^K$$

$$f(x) = w_0 + w_1\sin(x) + w_2\cos(x)$$

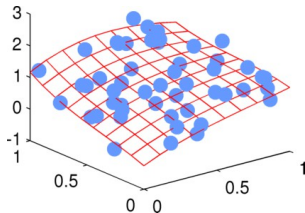
$$f(\mathbf{x}) = w_0 + w_1\sin(x_1) + w_2x_2^3$$

Example:

$$f(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2$$



$$f(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_2^2 + w_5x_1x_2 \\ + w_6x_1^3 + w_7x_1^2x_2 + w_8x_1x_2^2 + w_9x_2^3$$



Linear regression in vector notation

The linear model can be written compactly using vector notation

$$\begin{aligned}f(\mathbf{x}; \mathbf{w}) &= w_0 + w_1x_1 + w_2x_2 + \cdots + w_Mx_M \\&= \sum_{j=0}^M w_jx_j \\&= \tilde{\mathbf{x}}^\top \mathbf{w} \\&= \mathbf{w}^\top \tilde{\mathbf{x}}\end{aligned}$$

where $\tilde{\mathbf{x}} = [x_0, x_1, \dots, x_M]^\top$ with $x_0 = 1$.

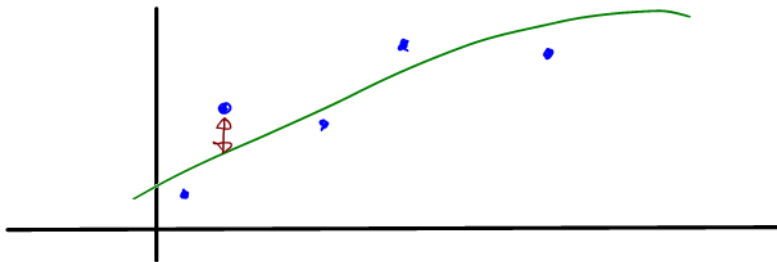
Evaluating a regression model

Compute average loss per observation:

$$E(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N L_p(y_i, f(\mathbf{x}_i; \mathbf{w}))$$

Where we either use L_1 or L_2 (Euclidean) loss

$$L_1(y_i, \hat{y}_i) = |y_i - \hat{y}_i|, \quad L_2(y_i, \hat{y}_i) = (y_i - \hat{y}_i)^2 \quad (\text{Compare these to } p\text{-norms})$$



Learning - the **classic** Least Squares solution

Setup: We got some data (y, X) . We have a way to define a function

$$f(x; w) = w^\top \tilde{x} = w_0 + \sum_{j=1}^M x_j w_j$$

Measuring the discrepancy and defining the loss function $E(w)$ function using the squared loss:

$$E(w) = \frac{1}{N} \sum_{i=1}^N (y_i - f(x_i; w))^2 = \frac{1}{N} \|y - \tilde{X}w\|_2^2 \quad (\text{aka RSS})$$

Learning (via optimization): w

$$\text{Optimal } w^* = \arg \min_w E(w)$$

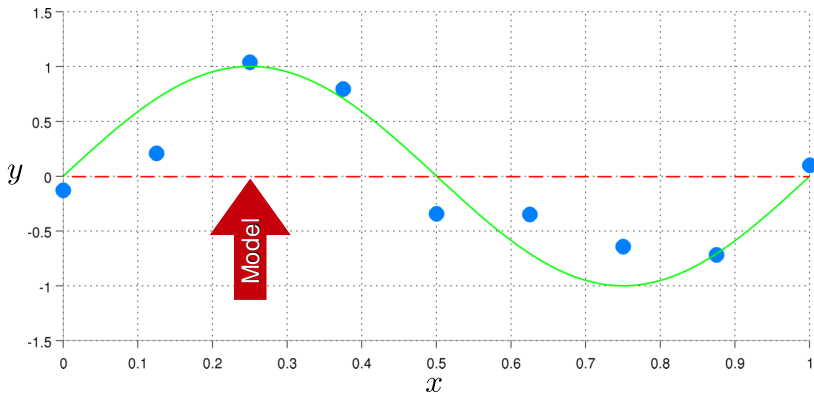
Solution:

$$\begin{aligned} \frac{\partial E(w)}{\partial w} &= 2\tilde{X}^\top (y - \tilde{X}w) = 0 \Rightarrow \tilde{X}^\top \tilde{X}w = \tilde{X}^\top y \Rightarrow \\ &\Rightarrow w^* = (\tilde{X}^\top \tilde{X})^{-1} \tilde{X}^\top y \end{aligned}$$

Linear regression in machine learning



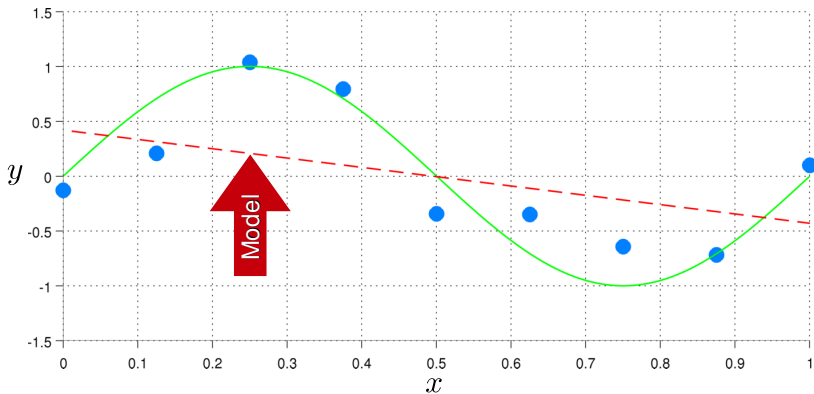
Linear regression



Model:

$$f(x) = w_0$$

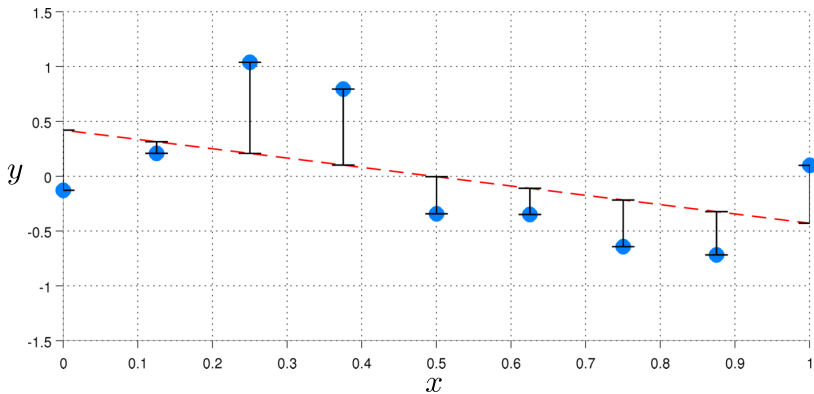
Linear regression



Model:

$$f(x) = w_0 + w_1x$$

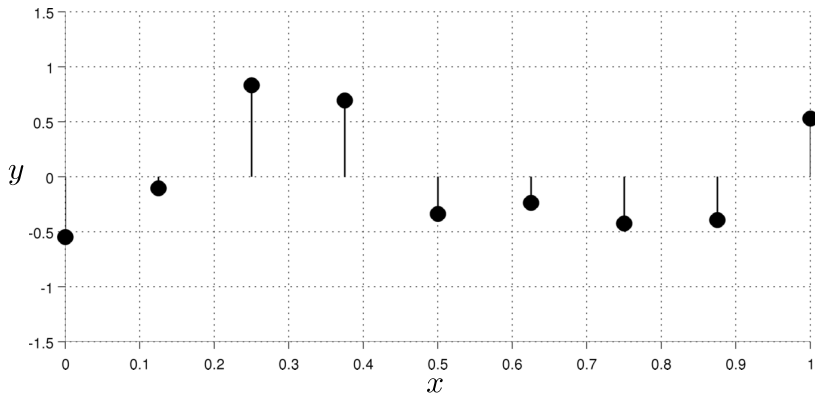
Linear regression—residual error



Model:

$$f(x) = w_0 + w_1x$$

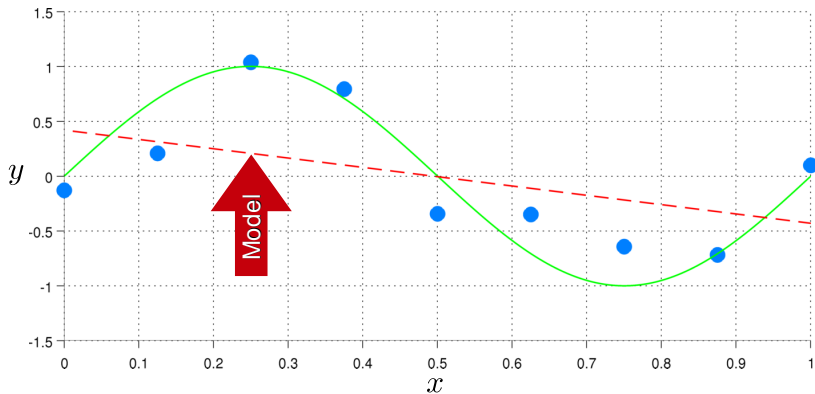
Linear regression—residual error



Model:

$$f(x) = w_0 + w_1x$$

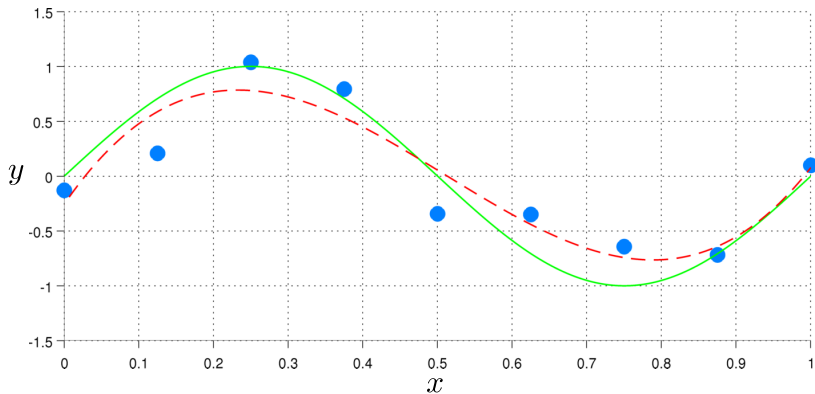
Linear regression



Model:

$$f(x) = w_0 + w_1x$$

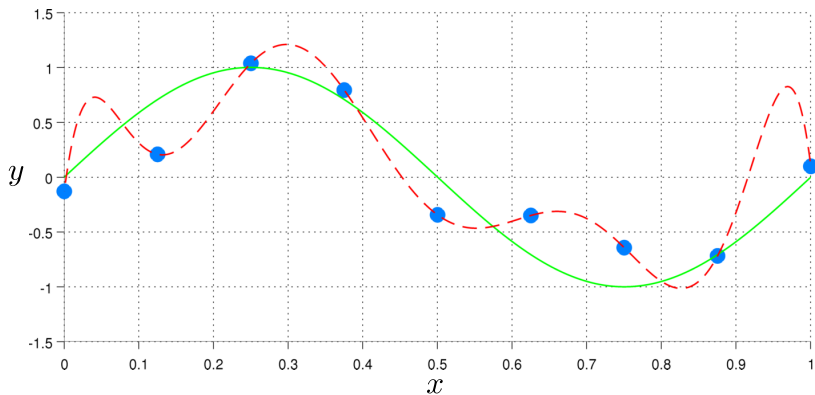
Linear regression



Model:

$$f(x) = w_0 + w_1x + w_2x^2 + w_3x^3$$

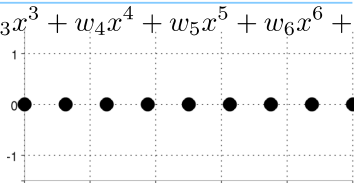
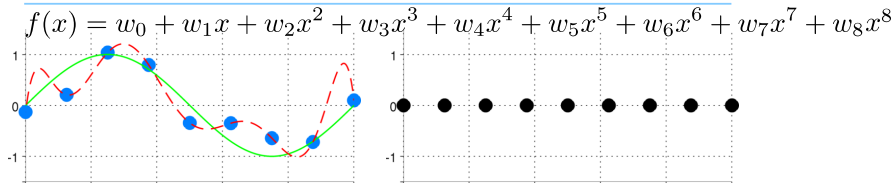
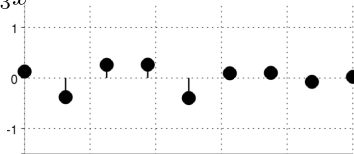
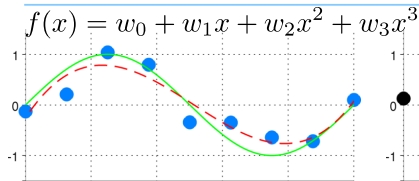
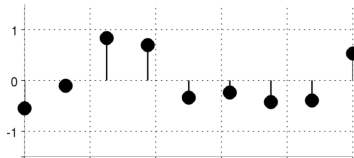
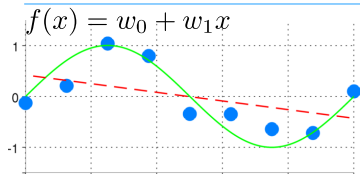
Linear regression



Model:

$$f(x) = w_0 + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5 + w_6x^6 + w_7x^7 + w_8x^8$$

Linear regression — model order



Learning

Setup: We got some data (y, X) . We have a way to define a function

$$f(x; w) = w^\top \tilde{x} = w_0 + \sum_{k=1}^M x_k w_k$$

- **Question:** How do we learn the correct w ?
- **Answer:** For each observation, assume:

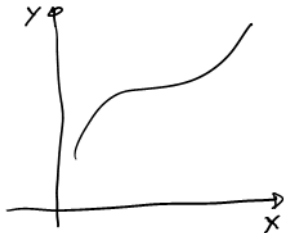
$$y_i = f(x_i; w) + \varepsilon_i$$

where ε_i is a normally distributed noise term $\mathcal{N}(\varepsilon_i | \mu = 0, \sigma^2)$. Recall:

$$\mathcal{N}(\varepsilon | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(\varepsilon - \mu)^2}{2\sigma^2}}$$

- This means that (since $\varepsilon_i = y_i - f(x_i, w)$)

$$p(y_i | x_i, w) = p(\varepsilon_i | \tilde{x}_i, w) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{([y_i - f(x_i, w)] - 0)^2}{2\sigma^2}} = \mathcal{N}(y_i | \underbrace{f(x_i, w)}_{\mu}, \sigma^2)$$



Recall: Maximum A Posteriori (MAP)

- Consider some data $\mathbf{X} = \mathbf{x}_1, \dots, \mathbf{x}_N$ and $\mathbf{y} = y_1, \dots, y_N$
- Suppose $\mathbf{x}_i$ relates to y_i by some parameters $\mathbf{w}$
- **Assume**

$$p(\mathbf{y}|\mathbf{X}, \mathbf{w}) = \prod_{i=1}^N p(y_i|\mathbf{x}_i, \mathbf{w}), \quad p(\mathbf{w}|\mathbf{X}) = p(\mathbf{w})$$

- Then

$$p(\mathbf{w}|\mathbf{X}, \mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{X}, \mathbf{w})p(\mathbf{w}|\mathbf{X})}{p(\mathbf{y}|\mathbf{X})} = \frac{\prod_{i=1}^N p(y_i|\mathbf{w}, \mathbf{x}_i)p(\mathbf{w})}{p(\mathbf{y}|\mathbf{X})}$$

- And maximizing: $\mathbf{w}^* = \arg \max_{\mathbf{w}} p(\mathbf{w}|\mathbf{X}, \mathbf{y})$ is equivalent to

$$\text{Minimize: } \mathbf{w}^* = \arg \min_{\mathbf{w}} E(\mathbf{w})$$

$$E(\mathbf{w}) = \frac{1}{N} \left[- \sum_{i=1}^N \log p(y_i|\mathbf{x}_i, \mathbf{w}) - \log p(\mathbf{w}) \right]$$

Back to the linear model

Likelihood:

$$p(y_i | \mathbf{x}_i, \mathbf{w}) = \mathcal{N}(y_i | f(\mathbf{x}_i, \mathbf{w}), \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - f(\tilde{\mathbf{x}}_i; \mathbf{w}))^2}{2\sigma^2}}$$

Optimal $\mathbf{w}^* = \arg \max_{\mathbf{w}} p(\mathbf{w} | \mathbf{X}, \mathbf{y})$ found as $\mathbf{w}^* = \arg \min_{\mathbf{w}} E(\mathbf{w})$

$$E(\mathbf{w}) = \frac{1}{N} \left[- \sum_{i=1}^N \log p(y_i | \mathbf{x}_i, \mathbf{w}) - \log p(\mathbf{w}) \right]$$

By assuming a constant (flat prior) we can ignore $p(\mathbf{w})$ and obtain (i.e., **MLE**)

$$E(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N \frac{(y_i - \mathbf{x}_i^\top \mathbf{w})^2}{2\sigma^2} = \frac{1}{N} \frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \tilde{\mathbf{x}}_i^\top \mathbf{w})^2 \propto \|\mathbf{y} - \tilde{\mathbf{X}}\mathbf{w}\|^2$$

$$\frac{\partial E(\mathbf{w})}{\partial \mathbf{w}} = 2\tilde{\mathbf{X}}^\top (\mathbf{y} - \tilde{\mathbf{X}}\mathbf{w}) = 0$$

$$\Rightarrow \mathbf{w}^* = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \mathbf{y}$$

• same as the
OLS view
!

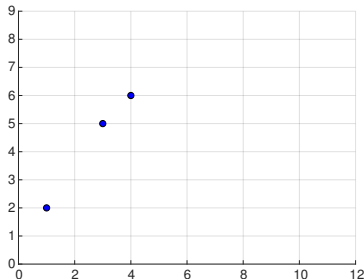
Quiz 1: The linear model

Suppose you observe three points:

$$(x, y) = (1, 2), (3, 5), (4, 6)$$

Knowing what you have learned so far, you first bring these points to the standard format:

$$\mathbf{X} = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 2 \\ 5 \\ 6 \end{bmatrix}$$



You wish to train a linear model of the form $y = ax + b$ on this dataset. What is $\mathbf{w} = \begin{bmatrix} b \\ a \end{bmatrix}$? Then, compute the prediction of the model at $x = 5$? (the prediction is given as: $y = \tilde{\mathbf{x}}^\top \mathbf{w}^*$)

- A. 6.5
- B. 7
- C. 7.5
- D. 8
- E. Don't know.

$$\text{Recall } \mathbf{w}^* = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \mathbf{y}$$

Pen & paper

- ① realize you need $\tilde{x}$ not x
(ie add const. 1 to x)

$$\tilde{x}_{\text{test}} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

$$\tilde{x} = \begin{bmatrix} 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$$

- ② Find w^*

$$a) \tilde{x}^T \tilde{x} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 8 & 26 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$$

- b) use that inverse of

$$\underline{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ is } \underline{A}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \text{ so } (\tilde{x}^T \tilde{x})^{-1} = \frac{1}{3 \cdot 26 - 8 \cdot 8} \begin{bmatrix} 26 & -8 \\ -8 & 3 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1.8571 & -0.5714 \\ 0.5714 & 0.2143 \end{bmatrix}$$

The solution is found by first computing w using the standard formula, and remembering to add a column of ones to X to account for the offset. We

get:

$$w \approx \begin{bmatrix} 0.7143 \\ 1.3571 \end{bmatrix}$$

Evaluating the model gives $f(5) = y = 7.5$.

$$\textcircled{3} \tilde{x}^T y = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 13 \\ 41 \\ 25 \\ 6 \end{bmatrix}$$

$$\textcircled{4} w^* = \begin{bmatrix} 1.8571 & -0.5714 \\ 0.5714 & 0.2143 \end{bmatrix} \begin{bmatrix} 1.8571 \cdot 13 - 0.5714 \cdot 41 \\ 0.5714 \cdot 13 + 0.2143 \cdot 41 \end{bmatrix} \approx \begin{bmatrix} 0.715 \\ 1.3591 \end{bmatrix}$$

$$\begin{bmatrix} 13 \\ 41 \end{bmatrix}$$

$$\textcircled{5} \text{ predict for } \tilde{x}_{\text{test}} \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

$$y_{\text{test}} = \begin{bmatrix} 0.715 & 1.3581 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \end{bmatrix} \approx 7.5$$

Classification

- **Definition:** Learning a function that maps a data object to a discrete class
- **Why classify?**
 - Descriptive modeling
 - Explain / understand the relation between attributes and class
 - Predictive modeling
 - Predict the class of a new data object

Logistic regression (linear models for classification)

- Assume we are given $(\mathbf{X}, \mathbf{y})$, but assume y is *binary*: $y_i \in \{0, 1\}$
- An idea is to use the Bernoulli distribution

$$p(y_i | \mathbf{x}_i, \mathbf{w}) = \text{Bernoulli}(y_i | \theta_i) = \theta_i^{y_i} (1 - \theta_i)^{1-y_i}$$

Where θ_i depends on $\mathbf{w}$ and $\mathbf{x}_i$.

- **Problem:** θ_i must belong to the unit interval $[0, 1]$, but $f(\mathbf{x}_i; \mathbf{w}) = \tilde{\mathbf{x}}_i^\top \mathbf{w} \in \mathbb{R}$
- **Solution:** Assume

$$\theta_i = \sigma(f(\mathbf{x}_i; \mathbf{w})), \quad \text{where } \sigma(z) = \frac{1}{1 + e^{-z}} \text{ is the } \mathbf{logistic\ sigmoid}$$

Then

$$\begin{aligned} -\log p(y_i | \mathbf{x}_i, \mathbf{w}) &= -\log [\text{Bern}(y_i | \theta_i = \sigma(\tilde{\mathbf{x}}_i^\top \mathbf{w}))] = -\log [\theta_i^{y_i} (1 - \theta_i)^{1-y_i}] \\ &= -y_i \log(\theta_i) - (1 - y_i) \log(1 - \theta_i) \end{aligned}$$

Recall: Maximum A Posteriori (MAP)

- Consider some data $\mathbf{X} = \mathbf{x}_1, \dots, \mathbf{x}_N$ and $\mathbf{y} = y_1, \dots, y_N$
- Assume**

$$p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) = \prod_{i=1}^N p(y_i \mid \mathbf{x}_i, \mathbf{w}), \quad p(\mathbf{w} \mid \mathbf{X}) = p(\mathbf{w})$$

- Then

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) = \frac{\prod_{i=1}^N p(y_i \mid \mathbf{x}_i, \mathbf{w}) p(\mathbf{w})}{p(\mathbf{y} \mid \mathbf{X})}$$

- Maximizing: $\mathbf{w}^* = \arg \max_{\mathbf{w}} p(\mathbf{w} \mid \mathbf{X}, \mathbf{y})$ is equivalent to $\mathbf{w}^* = \arg \min_{\mathbf{w}} E(\mathbf{w})$

$$E(\mathbf{w}) = \frac{1}{N} \left[- \sum_{i=1}^N \log p(y_i \mid \mathbf{x}_i, \mathbf{w}) - \log p(\mathbf{w}) \right]$$

- By assuming a constant (flat prior) we can ignore $p(\mathbf{w})$ and obtain (i.e., **MLE**)

$$E(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N [-y_i \log(\theta_i) - (1 - y_i) \log(1 - \theta_i)], \quad \theta_i = \sigma(\tilde{\mathbf{x}}_i^\top \mathbf{w}) = \frac{1}{1 + e^{-\tilde{\mathbf{x}}_i^\top \mathbf{w}}}$$

Quiz 2: Logistic regression

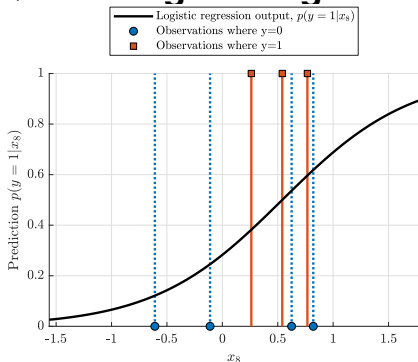


Figure 1: Output of a logistic regression classifier trained on 7 observations from the dataset.

Consider the Avila Bible dataset. We are particularly interested in predicting whether a bible copy was written by copyist 1, and we therefore wish to train a logistic regression classifier to distinguish between copyist one vs. copyist two and three.

To simplify the setup further, we select just 7

observations and train a logistic regression classifier using only the feature x_8 as input (as usual, we apply a simple feature transformation to the inputs to add a constant feature in the first coordinate to handle the intercept term). To be consistent with the lecture notes, we label the output as $y = 0$ (corresponding to copyist one) and $y = 1$ (corresponding to copyist two and three).

In Figure 1 is shown the predicted output probability an observation belongs to the positive class, $p(y = 1|x_8)$. What are the weights?

- A. $\begin{bmatrix} -0.93 \\ 1.72 \end{bmatrix}$
- B. $\begin{bmatrix} -2.82 \\ 0.0 \end{bmatrix}$
- C. $\begin{bmatrix} 1.36 \\ 0.4 \end{bmatrix}$
- D. $\begin{bmatrix} -0.65 \\ 0.0 \end{bmatrix}$
- E. Don't know.

Pen & paper solution (+ basic calculator with exp())

- ① $p(y|w, x) = \theta^y (1-\theta)^{1-y}$
- ② $y=1$:
 $p(y|w, x) = \theta$
 $\theta = \sigma(w^T \tilde{x})$
 $= \frac{1}{1 + e^{-(w_0 + w_1 x_8)}}$
- ③ let's try $x_8=1$:
 $\frac{1}{1 + e^{-(w_0 + w_1)}}$

we know from the plot that this should be ≈ 0.7 @ $x_8=1$
- ④ Try out the different options

A: $\frac{1}{1 + e^{-(0.43 + 1.72)}} \approx 0.69$ ✓

B: ≈ 0.06

C: ≈ 0.85

D: ≈ 0.34

Get to know $\sigma(z)$!!

The solution is easily found by simply computing the predicted $\hat{y} = p(y = 1|x_8)$ -value for an appropriate choice of x_8 . Notice that

$$p(y = 1|x_8) = \sigma(\tilde{x}_8^T w)$$

If we select $x_8 = 1$ and select the weights as in option

A we find $p(y = 1|x_8) = 0.69$, in good agreement with the figure. On the other hand, for the weights in option C we obtain $\hat{y} = 0.85$, for D that $\hat{y} = 0.34$ and finally for B that $\hat{y} = 0.06$. We can therefore conclude that A is correct.

Logistic regression: The optimization aspects*

- For linear regression, we derived a closed-form for $\mathbf{w}$
- For logistic regression, we have:

$$E(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N [-y_i \log(\theta_i) - (1 - y_i) \log(1 - \theta_i)], \quad \theta_i = \sigma(\tilde{\mathbf{x}}_i^\top \mathbf{w}) = \frac{1}{1 + e^{-\tilde{\mathbf{x}}_i^\top \mathbf{w}}}$$

- The gradient turns out to be easily written as (you do not need to know this):

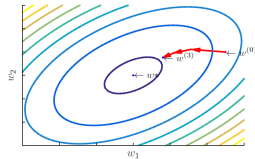
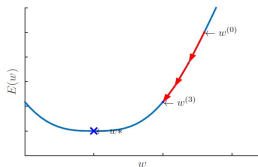
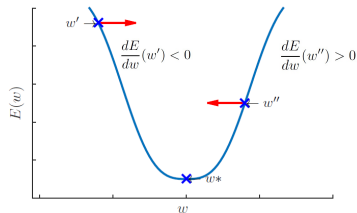
$$\frac{\partial E(\mathbf{w})}{\partial \mathbf{w}} = \frac{1}{N} \tilde{\mathbf{X}}^\top (\boldsymbol{\theta} - \mathbf{y}) = \frac{1}{N} \tilde{\mathbf{X}}^\top (\sigma(\tilde{\mathbf{X}}^\top \mathbf{w}) - \mathbf{y}).$$

We sometimes write this as $\nabla_{\mathbf{w}} E(\mathbf{w}) = \left[\frac{\partial E(\mathbf{w})}{\partial w_1} \quad \frac{\partial E(\mathbf{w})}{\partial w_2} \quad \dots \quad \frac{\partial E(\mathbf{w})}{\partial w_d} \right]^\top$

Logistic regression: The optimization aspects*

- Start from an initial guess $w^{(0)}$ for the optimal w^*
- At step t of the algorithm, modify $w^{(t)}$ to produce a better guess $w^{(t+1)}$

$$w^{(t+1)} = w^{(t)} - \eta \frac{dE}{dw}(w^{(t)})$$



Logistic regression: A gradient descent approach*

A simple optimization algorithm:

- 1 Initialize parameters: Choose an initial guess $\mathbf{w}^{(0)}$ (e.g., random values or zeros).
- 2 Choose a learning rate. Pick a step size $\eta > 0$ (the learning rate).
- 3 Iterate until convergence (for $t = 0, 1, 2, \dots$)
 - 3.a Compute the gradient: $\nabla_{\mathbf{w}} E(\mathbf{w}^{(t)})$
 - 3.b Update the parameters by stepping *against* the gradient:
$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \nabla_{\mathbf{w}} E(\mathbf{w}^{(t)})$$
- 4 Stopping criteria Stop when small gradient norm: $\|\nabla E(\mathbf{w}^{(t)})\| < \varepsilon$, tiny improvement in loss: $|E(\mathbf{w}^{(t+1)}) - E(\mathbf{w}^{(t)})| < \varepsilon$ or maximum number of iterations reached.

Generalized linear model

$$\text{Linear regr.: } E(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N \|y_i - \tilde{\mathbf{x}}_i^\top \mathbf{w}\|^2$$

$$\text{Logistic regr.: } E(\mathbf{w}) = \frac{-1}{N} \sum_{i=1}^N [y_i \log(\theta_i) + (1 - y_i) \log(1 - \theta_i)], \quad \theta_i = \sigma(\tilde{\mathbf{x}}_i^\top \mathbf{w})$$

$$\text{GLM: } E(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N d(y_i, g(\tilde{\mathbf{x}}_i^\top \mathbf{w}))$$

We call d the cost function and g the link function. In our examples:

$$\text{Lin.reg. : } d(y, z) = \|y - z\|^2, \quad z = g(\tilde{\mathbf{x}}_i^\top \mathbf{w}) = \tilde{\mathbf{x}}_i^\top \mathbf{w}$$

$$\text{Log.reg. : } d(y, z) = -y \log z - (1 - y) \log(1 - z), \quad z = g(\tilde{\mathbf{x}}_i^\top \mathbf{w}) = \sigma(\tilde{\mathbf{x}}_i^\top \mathbf{w})$$

Summary

Summary

- **Learning principles** via probabilistic inference
 - Introduction to probabilistic (/Bayesian) inference
 - Maximum likelihood estimation (MLE)
 - Maximum a posteriori (MAP)
- **Likelihoods** $p(y | x)$ is at the core and allows us to form principled loss/error functions
 - Discrete random variables and related distributions
 - Bernoulli distribution (for classification)
 - *Later*: Categorical distribution
 - Continuous random variables and related distributions
 - Multivariate normal distribution (aka multivariate Gaussians)
 - Beta distribution
 - *Later*: Mixture of multivariate normal distributions (aka Gaussian mixture model)
- **Linear models** to parametrise the likelihood (e.g. the mean or probability of a class via the sigmoid)

Resources

DTU Compute Our interactive regression demo

(<http://www2.imm.dtu.dk/courses/02450/DemoComplexityRegression.html>)

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